

Coding & Solution

Date	11 July 2026
Team ID	xxxxxx
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mallelauday/StudyAI-platform.git
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	React.js, Vite, Flask, Tailwind CSS
Key Features Implemented	User Authentication; Study Material Upload (PDF, DOCX, TXT); AI Summary Generator; AI Flashcards; AI Quiz Generator (MCQ, True/False, Short Answer); Personalized Study Plans; Learning Analytics Dashboard; Weak Topic Analysis; Firebase Integration; Local JSON Fallback; Responsive UI
Pending / Incomplete Features	Email Notifications; OCR Support for Scanned PDFs; Multi-language Support; Offline Mode; Real-time Collaboration; Mobile Application
Setup / Run Instructions	1. Clone the repository. 2. Install frontend and backend dependencies. 3. Configure the .env file with the Groq API key and Firebase credentials. 4. Run the Flask backend using "python app.py". 5. Run the React frontend using "npm install" followed by "npm run dev"

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments /	Yes

	documentation where necessary	
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The StudyAI platform follows a modular full-stack architecture using React.js for the frontend and Flask for the backend. Firebase Firestore is used as the primary database with a Local JSON fallback mechanism for offline development. The application integrates the Groq API (Llama 3.3-70B Versatile) for AI-powered summarization, flashcard generation, quiz creation, personalized study planning, and learning analytics. The codebase follows clean coding practices, RESTful API design, environment-based configuration, and Git-based version control to support maintainability and future scalability.